Problem 1

u_{ij} : Units of product i produced on Machine j.

c_{ij} : The time cost of producing product i on Machine j.

p_i : The profit of product i.

∴ The decision variable:

u_{A1} : Units of product A produced on Machine 1

u_{B1} : Units of product B produced on Machine 1

u_{C1} : Units of product C produced on Machine 1

u_{A2} : Units of product A produced on Machine 2

u_{B2} : Units of product B produced on Machine 2

u_{C2} : Units of product C produced on Machine 2

y : an auxiliary variable $\quad y = \begin{cases} 0, & u_{A1}+u_{A2}+u_{B1}+u_{B2}+u_{C1}+u_{C2} \le 20 \\ 1, & \text{otherwise} \end{cases}$

Objective function:

$$\text{maximize} \quad 30(u_{A1}+u_{A2}) + 40(u_{B1}+u_{B2}) + 50(u_{C1}+u_{C2}) - 200y$$
$$u_{A1}, u_{A2}, u_{B1}, u_{B2}, u_{C1}, u_{C2}$$

Constraints:

$u_{A1}+u_{A2} \ge 8$

$u_{B1}+u_{B2} \ge 10$

$u_{C1}+u_{C2} \ge 5$

$3u_{A1}+2u_{B1}+u_{C1} \le 60$

$2u_{A2}+4u_{B2}+3u_{C2} \le 72$

$u_{A1}+u_{A2}+u_{B1}+u_{B2}+u_{C1}+u_{C2} \le 40$

$u_{A1}+u_{A2} \le 0.7 \times (u_{A1}+u_{A2}+u_{B1}+u_{B2}+u_{C1}+u_{C2})$

$u_{B1}+u_{B2} \le 0.7 \times (u_{A1}+u_{A2}+u_{B1}+u_{B2}+u_{C1}+u_{C2})$

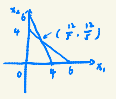
$u_{C1}+u_{C2} \le 0.7 \times (u_{A1}+u_{A2}+u_{B1}+u_{B2}+u_{C1}+u_{C2})$

$u_{A1}, u_{A2}, u_{B1}, u_{B2}, u_{C1}, u_{C2} \in \mathbb{Z}$

$y \in \{0,1\}$

Problem 2

$$\max \quad 3x_1 + 4x_2$$
$$\text{s.t.} \quad 2x_1 + 3x_2 \le 12$$
$$3x_1 + 2x_2 \le 12$$
$$x_1, x_2 \ge 0, \quad x_1, x_2 \in \mathbb{Z}$$



∴ The optimal solution for LP relaxation is $(\frac{12}{5}, \frac{12}{5})$ graphically

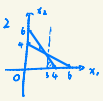
∴ S1 : Add the constraint $x_1 \le 2$

S2 : Add the constraint $x_1 \ge 3$

∴ S1 :



S2



∴ The optimal solution for S1 LP relaxation is $(2, \frac{8}{3})$, with optimal value $\frac{50}{3}$

The optimal solution for S2 LP relaxation is $(3, \frac{3}{2})$, with optimal value 15

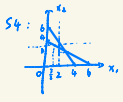
∴ S3 : Based on S1, add a constraint $x_2 \le 2$

S4 : Based on S1, add a constraint $x_2 \ge 3$

∴ S3 :



S4 :



∴ S3 : Optimal solution is $(2,2)$ with optimal value 14

S4 : Optimal solution is $(\frac{1}{2}, 3)$ with optimal value 16.5

A feasible solution in S4 : $(0,4)$, with value $16 > 15$

∴ S2 could be cut.

∴ Divide S4 into S5, S6

S5 : add $x_1 \le 1$ S6 : add $x_1 \ge 2$

∵ $x_1 \le 2$ ∴ For S6 : $x_1 = 2$, $x_2 \ge 3$ ⇒ infeasible.

For S5 : The Optimal solution is $(0,4)$, with optimal value 16

∴ In summary, the optimal solution is $(0,4)$ with optimal value 16.